Self-Check

9. Which tool could you use to determine if a laptop's power cable is working properly?
   a. Continuity tester
   b. Cable performance tester
   c. Toner and probe kit
   d. Multimeter
10. Which tool could you use to test a twisted-pair cable's pinout?
   a. OTDR
   b. Toner probe
   c. OPM
   d. Continuity tester

☐ Check your answers at the end of this chapter.

Summary

Section 5-1: Transmission Basics

- Frequency is typically measured in MHz (megahertz) or GHz (gigahertz), which indicates the number of times in a second that an electrical signal can change states (for example, change from a positive to a negative charge or vice versa). Bandwidth is measured in Mbps (megabits per second) or Gbps (gigabits per second) and refers to the amount of data you could theoretically transmit during a given period of time, taking into consideration factors such as frequency, distance, and SNR (signal-to-noise ratio). Throughput (also called payload rate or effective data rate) is also measured in Mbps (megabits per second) or Gbps (gigabits per second) and refers to the number of data bits (0s and 1s) that are actually received across a connection each second. Throughput takes into consideration the reality of a network environment.
- Noise, or interference, can degrade or distort a signal and, on a network, is measured in dB (decibels). Crosstalk occurs when a signal traveling on one wire or cable infringes on the signal traveling over an adjacent wire or cable.
- Attenuation is the loss of a signal's strength as it travels away from its source. To compensate for attenuation, signals are boosted en route using a repeater, which regenerates a digital signal in its original form without the noise it might have previously accumulated.
- Latency is the brief delay between the instant when data leaves the source and when it arrives at its destination. The most common way to measure latency on data networks is by calculating a packet's RTT (round trip time), or the length of time it takes for a packet to go from sender to receiver, then back from receiver to sender. If packets experience varying amounts of delay, they can arrive out of order—a problem commonly called jitter, or more precisely, PDV (packet delay variation).
- Fiber-optic cables typically use WDM (wavelength division multiplexing), which divides a light beam into different wavelengths, DWDM (dense WDM), which increases the number of channels provided by normal WDM, and CWDM (coarse WDM), which lowers cost by spacing frequency bands wider apart to allow for cheaper transceiver equipment.

Section 5-2: Copper Cable

- Coaxial cable, called "coax" for short, was the foundation for Ethernet networks in the 1980s. RG-6 coax is still used for cable Internet, cable TV, and some multimedia connection types. An F-type connector attaches to coaxial cable so that the pin in the center of the cable extends into the center of the connector. A BNC connector connects to another BNC connector via a turn-and-lock mechanism.
- Twinaxial cable, called "twinax" for short, looks very similar to coax cable except that there are two cores, or conductors, inside the cable. More recent twinax cables contain multiple pairs of these cores to carry even more data. Twinax is an inexpensive option for short, high-speed connections, such as when connecting switches to routers or servers in a data center. For this reason, twinax is also called a DAC (direct attach copper) cable.
- Twisted-pair cabling in Ethernet networks contains four wire pairs. On Fast Ethernet networks, which have a maximum speed of 100 Mbps, one pair sends data, another pair receives data, and the other two pairs are not used for data transmission. Networks using Gigabit Ethernet and higher standards, with a speed of at least 1,000 Mbps, use all four pairs for both sending and receiving.

- Modern LANs use Cat 5e or higher wiring, which is the minimum required to support Gigabit Ethernet. Cat 6 and above are certified for multigigabit transmissions, although Cat 6 cable has shorter distance limitations when supporting 10 Gbps. While Cat 7/7a cables never gained significant popularity, Cat 8 cables are already widely available even for home networks.
- STP (shielded twisted pair) cable consists of twisted-pair wires that are not only individually insulated, but might also be surrounded by a shielding made of a metallic substance such as a foil. The shielding acts as a barrier to external electromagnetic forces, thus preventing them from affecting the signals traveling over the wire inside the shielding. UTP cabling does not contain additional shielding for the twisted pairs. As a result, UTP is both less expensive and less resistant to noise than STP.
- TIA/EIA has specified two methods of inserting twisted-pair wires into RJ-45 plugs: TIA/EIA-568A and TIA/EIA-568B (also known as T568A and T568B, respectively). Functionally, there is very little difference between these two standards. You only must be certain that you use the same standard on every RJ-45 plug and jack on your network so data is transmitted and received correctly.
- The most common type of networking cable is a straight-through cable, also called a patch cable. To create one, terminate the RJ-45 plugs at both ends of the cable identically, following one of the TIA/EIA-568 standards.
- PoE (Power over Ethernet) is a method for supplying electrical power over twisted-pair Ethernet connections. The amount of power provided is relatively small—15.4 watts for standard PoE devices and 25.5 watts for the newer PoE+ devices.
- A device's NIC is rated for maximum network speeds, which are defined by various Ethernet standards. When two devices first connect, they auto-negotiate for the fastest standard they have in common, such as 100BASE-TX, 1000BASE-T, or 10GBASE-T.

Section 5-3: Fiber-Optic Cable

- Fiber-optic cable, or simply "fiber," contains one or several glass or plastic fibers at its center, or core. Data is transmitted through the central fibers via pulsing light typically sent from a laser or an LED (light-emitting diode). SMF (single mode fiber) consists of a narrow core of 8 to 10 microns in diameter. Laser-generated light travels a single path over the core, reflecting very little.
- MMF (multimode fiber) contains a core with a larger diameter than SMF, usually 50 or 62.5 microns, over which many pulses of light generated by a laser or LED light source travel at various angles. Signals traveling over multimode fiber experience greater attenuation than those traversing single mode fiber. Therefore, MMF is not suited to distances longer than a few kilometers.
- The transition between SMF and MMF cabling might occur at an FDP (fiber distribution panel), which is usually a case on a rack where fiber cables converge, connect with each other, and connect with fiber-optic terminal equipment from the ISP.
- SMF connectors are typically available with a 1.25-mm ferrule or a 2.5-mm ferrule, though other sizes can be found. The most common 1.25-mm ferrule connector is LC (local connector). Two 2.5-mm ferrules are SC (subscriber connector or standard connector) and ST (straight tip). A common type of MMF connector is MT-RJ (mechanical transfer-registered jack). A newer connector that supports both MMF and SMF is MPO (multi-fiber push on).
- Common fiber transceivers include SFP (small form-factor pluggable), SFP+, QSFP (quad small form-factor pluggable), and QSFP+. SFP was the first to offer Gigabit Ethernet speeds with SFP+ improving on those speeds. QSFP was the first to squeeze four channels in a single transceiver to support significantly higher data rates.
- IEEE developed standards for achieving high data rates over fiber-optic cable, which is the best medium for delivering high throughput. Common fiber Ethernet standards include 1000BASE-LX, 1000BASE-SX, 10GBASE-SR, and 10GBASE-LR.

Section 5-4: Cable Troubleshooting Tools

- A steady network connection LED status indicator light on a network port indicates connectivity, and a blinking light indicates activity. A red or amber light might indicate a problem.
- A toner and probe kit can help determine where a wire, possibly out of hundreds, terminates. Tone generators and tone locators cannot be used to determine any characteristics about a wire, such as whether it's defective or whether its length exceeds IEEE standards for a certain type of network.
- A multimeter is a simple instrument that can measure many characteristics of an electric circuit, including its resistance, voltage, and impedance. You might use a multimeter to measure voltage to verify that a cable is properly conducting electricity, to check for the presence of noise on a wire, or to test for short or open circuits in the wire.
- A continuity tester determines whether a cable is carrying a data signal to its destination. Some continuity testers will verify that the wires in a UTP or STP cable are paired correctly following TIA/EIA 568 standards and

that they are not shorted, exposed, or crossed. Continuity testers for fiber-optic cables issue light pulses on the fiber and determine whether they reach the other end of the fiber.

- A sophisticated cable performance tester measures the overall performance of a cabling structure. In addition to tests a continuity tester can perform, a cable performance tester can also be used to measure the distance to a connectivity device, termination point, or damage in a cable; measure attenuation along a cable; measure NEXT (near end crosstalk) between wires as well as alien crosstalk; measure termination resistance and impedance; issue pass/fail ratings for various categories of cabling standards; store and print cable testing results or directly save data to a computer database; and graphically depict a cable's attenuation and crosstalk characteristics over the length of the cable.
- A sophisticated performance tester will include a TDR (time domain reflectometer). A TDR issues a signal on a cable and then measures the way the signal bounces back (or reflects) to the TDR. Performance testers for fiber-optic connections use OTDRs (optical time domain reflectometers) that transmit light-based signals of different wavelengths over the fiber.
- An OPM (optical power meter), also called a fiber light meter, measures the amount of light power transmitted on a fiber-optic line. The device must be calibrated precisely; however, the surrounding room temperature, connection type, and skill of the technician conducting the test all affect the accuracy of the final test results.

Key Terms

For definitions of key terms, check the Glossary.

1000BASE-LX
1000BASE-SX
1000BASE-T
100BASE-T
100BASE-TX
10BASE-T
10GBASE-LR
10GBASE-SR
10GBASE-T
40GBASE-T
802.3 standards
attenuation
auto-MDI-X
bandwidth
BNC (Bayonet Neill-Concelman) connector
cable crimper
cable diagram
cable map
cable performance tester
cable stripper
cable tester
Cat 5 (Category 5)
Cat 5e (Enhanced Category 5)
Cat 6 (Category 6)
Cat 6a (Augmented Category 6)
Cat 7 (Category 7)
Cat 7a (Augmented Category 7)
Cat 8 (Category 8)
CFP (centum form-factor pluggable)
cladding
coaxial cable
console cable
continuity
continuity tester
core
crossover cable
DAC (direct attach copper) cable
dB (decibel)
duplex
encoding
Ethernet standards
Fast Ethernet
FDP (fiber distribution panel)
ferrule
fiber light meter
frequency
F-type connector
full-duplex
fusion splicer
Gigabit Ethernet
impedance
interference
jitter
latency
LC (local connector)
LED (light-emitting diode)
loopback adapter
MDI-X (medium dependent interface crossover or MDI crossover)
MMF (multimode fiber)
modal bandwidth
modulation
MPO (multi-fiber push on)
MT-RJ (mechanical transfer-registered jack)
multimeter
multiplexing
open circuit
OPM (optical power meter)
optical link budget
optical loss
OTDR (optical time domain reflectometer)
patch cable
pinout
plenum-grade cable
PoE (Power over Ethernet)
PoE+
QSFP (quad small form-factor pluggable)
QSFP+
repeater
riser-rated cable
RJ-11 (registered jack 11)
RJ-45 (registered jack 45)
rollover cable
RTT (round trip time)
SC (subscriber connector or standard connector)
SFP (small form-factor pluggable)

SFP+
short circuit
signal degradation
SMF (single mode fiber)
snips
speed and duplex mismatch
ST (straight tip)
straight-through cable
TDR (time domain reflectometer)
throughput
TIA/EIA-568A
TIA/EIA-568B
tone generator (toner)
tone locator (probe)
transceiver
twinaxial cable
twist ratio
twisted-pair cable
TX/RX reverse
visual fault locator
wire cutter
wire map
XFP (10 Gigabit small form-factor pluggable)

Acronyms Checklist

The acronyms in Table 5-7 are listed in the CompTIA Network+ objectives and could appear on the Network+ exam. This means that exam questions might use any of these acronyms in context so that you must know the meaning of the acronym in order to answer the question correctly. Make sure you're familiar with what each acronym stands for and the general concept of the term itself. All these acronyms are used in context in this chapter.

Table 5-7 Chapter 5 acronyms

Acronym	Spelled out
BNC	Bayonet Neill-Concelman
DAC	direct attach copper
LC	local connector
MDI-X or MDIX	medium dependent interface crossover
MPO	multifiber push on
PoE	power over Ethernet
QSFP	quad small form-factor pluggable
RJ	registered jack
RX	receiver
SC	subscriber connector or standard connector
SFP	small form-factor pluggable
ST	straight tip
TX	transmitter

Review Questions

1. Which transmission characteristic is never fully achieved?
   a. Latency
   b. Throughput
   c. Bit rate
   d. Bandwidth
2. Which kind of crosstalk occurs when the insulation around two wire pairs is damaged near the source of the signal?
   a. Alien
   b. TX/RX reverse
   c. FEXT
   d. NEXT
3. Which cable is best suited for ultra-high-speed connections between a router and a switch on the same rack?
   a. RG-6 coaxial cable
   b. Passive twinaxial cable
   c. Cat 5e twisted-pair cable
   d. Cat 6 twisted-pair cable
4. Which of these categories of twisted-pair cable can support Gigabit Ethernet?
   a. Cat 5, Cat 6, Cat 7
   b. Cat 5e, Cat 6, Cat 3
   c. Cat 5e, Cat 6a, Cat 7
   d. Cat 6, Cat 7a, Cat 5

5. Suppose you're creating patch cables to be used in a government office. What color wire goes in the first pin?
   a. White/green
   b. White/orange
   c. Brown
   d. Blue
6. What is the earliest twisted-pair cabling standard that meets the minimum requirements for 10GBASE-T transmissions at 100 meters?
   a. Cat 5e
   b. Cat 6
   c. Cat 6a
   d. Cat 7
7. What type of fiber-cable problem is caused when pairing a 50-micron core cable with a 62.5-micron core cable?
   a. Dirty connectors
   b. Wavelength mismatch
   c. TX/RX reverse
   d. Fiber type mismatch
8. What characteristic of optical transmission is primarily responsible for the distance limitations of fiber-optic cabling?
   a. Optical loss
   b. Optical link budget
   c. Cladding
   d. Noise immunity
9. Why is SMF more efficient over long distances than MMF?
   a. The SMF core reflects more light.
   b. SMF light is generated by an LED light source.
   c. SMF is less expensive.
   d. SMF has a narrower core.
10. Which of the following statements about short circuits and open circuits is true?
   a. Missing needed connections is a short circuit.
   b. Missing needed connections is an open circuit.
   c. Having connections where they shouldn't be is an open circuit.
   d. Having connections where they should be is a short circuit.

Scenario-Based Questions

Question 5-1

As you're setting up a wireless connection across a sports field on a school campus, you need to measure the signal's loss of strength from one side to the other. The manufacturer documentation states that the connection can tolerate up to a 50 percent loss of strength and still support target data throughput. Given these parameters, which of the following measurements would be the maximum decibel loss you could accept?

a. –1 dB
b. –2 dB
c. –3 dB
d. –6 dB

Question 5-2

After some renovation work in your office building, you're troubleshooting a failed network connection between a data closet and the server room. You've determined the source of the problem is a damaged cable. To avoid tearing into walls unnecessarily, you need to pinpoint the location of the damage. What kind of tool can measure the distance to the location of damage in a cable?

a. Multimeter
b. Toner probe
c. Continuity tester
d. Line tester

Question 5-3

Suppose you're assisting with a cable installation using fiber-optic cabling that will support Gigabit Ethernet. You're approved to install segments up to 4,000 m in length. What mode of fiber cable are you using?

a. SMF
b. MMF
c. SFP
d. QSFP

Question 5-4

You're testing cables running into a data closet to determine which cable runs to a wall outlet in a nearby classroom. As you're using the toner and probe kit, which part of the device emits an audible tone when it detects electrical activity on a wire pair?

a. TDR
b. Tone generator
c. Tone locator
d. Toner

Hands-On Projects

Note 5-13 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Project 5-1: Latency Around the World

Estimated time: 20 minutes (+5 minutes for group work, if assigned)
Objective 5.4: Given a scenario, troubleshoot common performance issues.
Group work: This project includes enhancements when assigned as a group project.
Resources:
- Internet access

Description:
Recall that IP tracks the number of times a message jumps or hops from one router to another on its way to its destination. Each of these hops requires a tiny bit of time—the more routers a message must traverse, the longer it will take to reach its destination. In this chapter, you learned that latency is the measure of the delay caused by the time it takes messages to travel over network media from one place to another. This concept is easy to see in the real world, where it takes longer, for example, for you to travel across the country than it does to go down the street to the grocery store. Even though network messages travel much faster than a car or a jet plane, it still takes time for them to get from one place to another. And then the response must also travel across a similar distance and number of hops, resulting in a longer RTT (round trip time). To see how distance affects a message's RTT, complete the following steps:

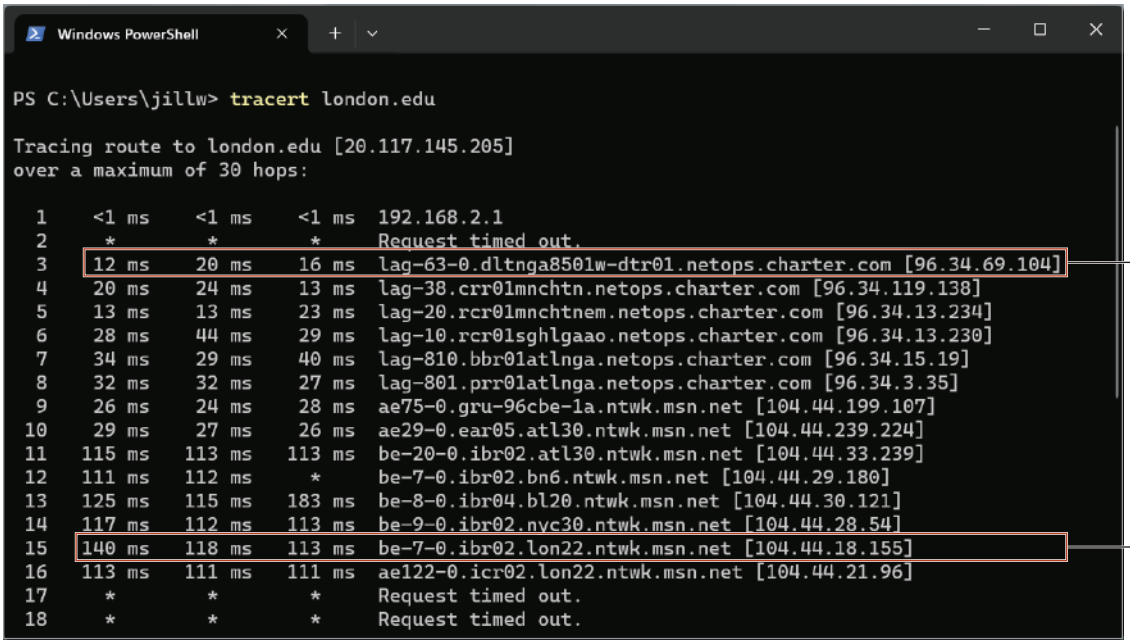
1. **For group assignments:** Open a PowerShell or Command Prompt window and run `tracert` on the IP address of one of your group members.
   a. If your group member is on the same LAN as you are, use their private IP address.
   b. If your group member is on a different network than you are, run `tracert` on their network's public IP address. You and they both can discover your respective network's public IP address using an IP address lookup tool online. Go to google.com and search for **What is my IP address**. The search results will list your network's public IP address at the top. Now that you know your public IP address, share that information with your group member and get their public IP address as well.
   c. How many hops did it take for your `tracert` messages to reach your group member's computer or network?

Note 5-14 | For an Ubuntu or other Linux installation, use traceroute rather than tracert for this project. You might need to first install the traceroute utility. On Ubuntu, run this command:

```
sudo apt install traceroute
```

2. In a PowerShell or Command Prompt window, run `tracert` on a website whose servers are located on a different continent from your location—across one ocean. For example, if you're located in the Midwest or Eastern United States, you can run the command `tracert london.edu` (London Business School). If you are on the West Coast, however, you might get more useful results for this step by targeting a server across the Pacific Ocean, such as `tracert www.tiu.ac.jp` (Tokyo International University). What command did you use?
3. Examine the output and find the point in the route where messages first exited your local network, and then find a point where messages started jumping across the ocean. By what percentage does the RTT increase after the jump across the ocean compared with before it? Consider the example in Figure 5-50.

Figure 5-50 The latency increases significantly as messages reach devices farther away



Source: Microsoft Corporation

To calculate the percentage of change in latency for the hops in Figure 5-50, select a time from after the jump across the ocean (140, for example) and divide it by a time from before the jump across the ocean (such as 20), then multiply by 100 percent: 140/20 × 100% = 700%. In this case, the sample data yields a 700 percent increase. It takes seven times as long for a message to go round-trip across the Atlantic from the United States to London, England (the location of this first European router), as it does for a message to travel round trip between two servers that are located near each other on the U.S. East Coast (this local computer and a U.S. router early in the route).

4. Choose a website whose servers are on a continent even farther away from you. For example, if you are in the United States, you could trace the route to the Indian Institute of Technology in India at the address **www.iitb.ac.in**. What command did you use? How many hops did it take until the route crossed an ocean? What other anomalies do you notice about this global route?

Note 5-15 | When performing a `tracert` on some websites, you might find the results show a final hop to AWS's Global Accelerator service (Figure 5-51). Global Accelerator is a cloud service that collects global traffic destined for an AWS customer's website and directs that traffic off the public Internet and, instead, across AWS's private infrastructure for reduced latency and improved user experience. While this result is especially helpful if you needed to interact with the target website, it's not particularly helpful when you're trying to compare latency measurements to various geographical locations around the world. To get more relevant results for this project, you should try again with a different website.

Figure 5-51 The final hop shows AWS's Global Accelerator service has taken over handling the traffic

```
Windows PowerShell

PS C:\Users\jillw> tracert du.ac.in

Tracing route to du.ac.in [3.33.160.68]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.2.1
  2     *        *        *     Request timed out.
  3    17 ms    34 ms    12 ms  lag-63-0.dltnga8501w-dtr01.netops.charter.com [96.34.69.104]
  4    11 ms    12 ms    12 ms  lag-38.crr01mnchtn.netops.charter.com [96.34.119.138]
  5    13 ms    24 ms    24 ms  lag-20.rcr01mnchtnem.netops.charter.com [96.34.13.234]
  6    28 ms    24 ms    23 ms  lag-10.rcr01sghlgaao.netops.charter.com [96.34.13.230]
  7    29 ms    26 ms    31 ms  lag-810.bbr01atlnga.netops.charter.com [96.34.15.19]
  8    31 ms    27 ms    24 ms  lag-801.prr01atlnga.netops.charter.com [96.34.3.35]
  9    23 ms    33 ms    25 ms  99.82.183.130
 10     *        *        *     Request timed out.
 11     *        *        *     Request timed out.
 12     *        *        *     Request timed out.
 13     *        *        *     Request timed out.
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16    25 ms    29 ms    24 ms  aeae1abfd9923c77d.awsglobalaccelerator.com [3.33.160.68]

Trace complete.
PS C:\Users\jillw>
```

AWS Global Accelerator service

Source: Microsoft Corporation

5. Choose one more website as close to directly across the globe from you as possible. U.S. locations might want to use the University of Western Australia at **uwa.edu.au**. What command did you use? How many hops are in the route? If indicated in the results, did the route go east or west around the world from your location? How can you tell?
6. Scott Base in Antarctica runs several webcams from various research locations. Run a trace to the Scott Base website at **antarcticanz.govt.nz**. What's the closest router to Scott Base's web server that your trace reached? If you can't tell from the command output where the last response came from, go to **iplocation.net** in your browser. Enter the final hop's IP address to determine that router's location.
7. Think about other locations around the world that might be reached through an interesting geographical route, such as traversing a place you would like to visit or tapping routers in an exotic location. Find a website hosted in that location and trace the route to it. Which website did you target? Where is it located? What are some router locations along the route of your trace? **Take a screenshot** of the output for your trace; submit this visual with your responses to this project's questions.
8. Try the `ping` command on several of these same IP addresses. Did it work? Why do you think this is the case?

Project 5-2: Create a Loopback Plug

Estimated time: 30 minutes
Objective 5.2: Given a scenario, troubleshoot common cabling and physical interface issues.
Resources:
- 6-8-inch length of UTP cabling (Cat 5 or Cat 5e)
- Unused RJ-45 plug
- Wire cutters, snips, or heavy-duty scissors
- Cable crimper
- A device (such as a desktop or laptop computer or a SOHO router) with a working Ethernet port that uses LED indicator lights

Note 5-16 | This hardware can be purchased in bulk and distributed to students. Alternatively, students can purchase their own supplies at stores such as Lowe's, Home Depot, Amazon.com, or Newegg.com.

Description:
In this chapter, you practiced terminating an Ethernet cable by attaching an RJ-45 connector. You also learned that a loopback plug crosses the transmit line with the receive line to trick a device into thinking it's connected to a network. You can create your own loopback plug by altering the pinout on the connector and forcing the transmissions to loop back in on themselves. A loopback plug is helpful for determining if a NIC on a workstation or a port on a switch is working. Complete the following steps:

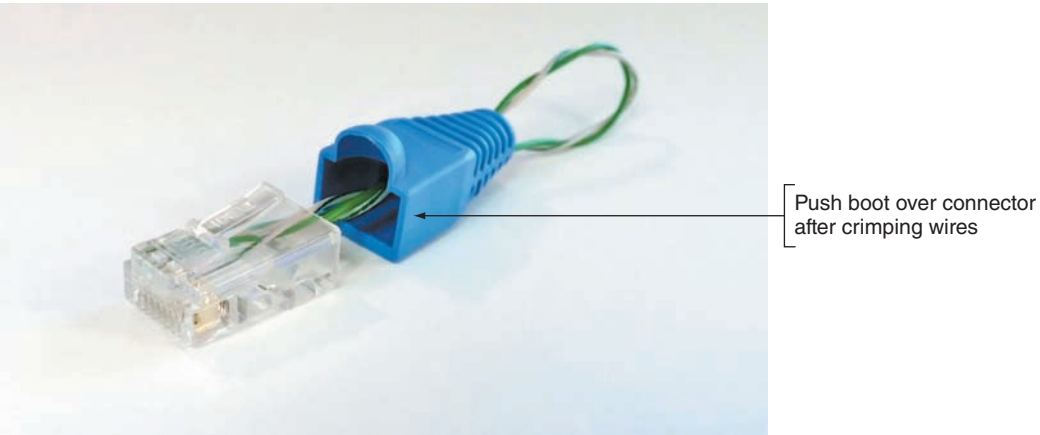
1. Cut to loosen the cable's covering, then slide the covering off the cable to separate the wire pairs into four groups. Flatten the wire pairs but do not untwist them. Select one wire pair (one solid and one striped) and lay the other pairs aside because you won't need them. Which wire pair did you choose?

2. Untwist the wires on each end an inch or less and straighten the tips. If needed, give each wire a clean cut to make sure the two wires on each end are even with each other.
3. Insert one end of the twisted pair into the RJ-45 plug, making sure the solid color wire goes into slot 1, and the striped wire goes into slot 2. Push the wires all the way into the slots. Make sure the wire tips touch the plastic surface at the front end inside the plug.
4. Loop the wire pair around and insert the other end into the plug. The solid color wire goes into slot 3, and the striped wire goes into slot 6. (Slots 4, 5, 7, and 8 are not needed unless you're testing Gigabit Ethernet equipment.)

Note 5-17 | If you want to include the other two pins in the adapter so you can test VoIP and similar Gigabit Ethernet equipment, you'll need to use a second twisted pair from your original cable. Before crimping, insert one end of the second pair into the plug. Press the solid color wire into slot 4 and the striped wire into slot 5. Loop the wire around and press the solid color wire into slot 7 and the striped wire into slot 8.

5. Push the wires all the way in and then use the crimper to secure the wires in the plug. If a boot came with the plug, you can insert it over the wire loop and push it all the way through to cover the wire/plug connection (Figure 5-52). **Take a photo of your loopback plug**; submit this visual with your responses to this project's questions.

Figure 5-52 Adding the boot to the loopback plug is optional



6. Insert the loopback plug into a device's Ethernet port that is known to be working correctly and has LED indicator lights. If the port's link indicator lights up (this might take a minute), you've successfully created a loopback plug.
7. Working with the actual hardware can be an enlightening experience as you work to get each wire lined up with the correct pin or realize how much force is required to crimp a cable inside the connector. What was the most difficult part of this project for you? What was the most satisfying part of the project?

Capstone Projects

Note 5-18 | Websites, applications, public cloud platforms, and related account options change often. While the instructions given in these projects were accurate at the time of writing, you might need to adjust the steps or options according to later changes.

Note to Instructors and Students: A rubric is provided for evaluating student performance on these projects. Please see Appendix D.

Capstone Project 5-1: Build a Packet Tracer Network

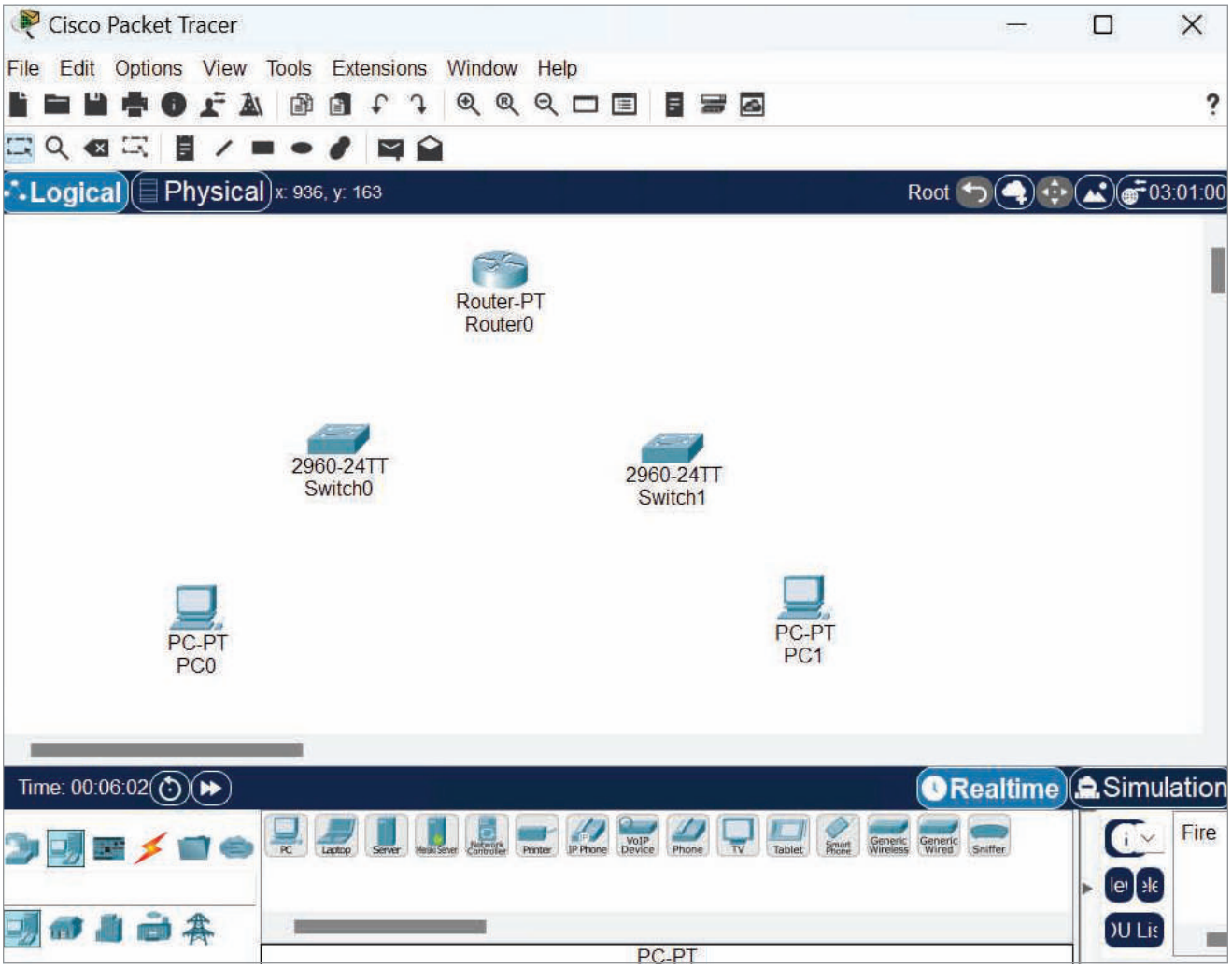
Estimated time: 45 minutes
Objective 1.7: Given a scenario, use appropriate IPv4 network addressing.
Resources:
- Computer with Cisco Packet Tracer installed
- Storage space for Packet Tracer network file, which will need to be accessed in later chapters

Description:
In earlier Capstone Projects, you have installed Packet Tracer and performed several kinds of tasks, such as working with MAC address tables and a TFTP server. Look back at your notes on your Wikidot website if you need help remembering details about what you learned in these earlier projects.

In this project, you will begin to build a more extensive network, and you'll continue building on this network in future chapters. Make sure you're working on a computer where you'll be able to save your Packet Tracer network file for later use, either by working on the same computer every time or by saving your file in the cloud where you can get to it later. The Packet Tracer network you begin building in this project is the starting point for many additions to this network in later chapters. To lay the foundation for working with your Packet Tracer network in several chapters, complete the following steps:

1. Open Cisco Packet Tracer and, if necessary, sign in with your Skills for All account.
2. From the Networking Devices > Routers group, add one **PT-Router** to the workspace.
3. From the Networking Devices > Switches group, add two **2960 switches** to the workspace.
4. From the End Devices group, add two **PCs** to the workspace.
5. Arrange these devices in a pyramid shape, with the workstations at the bottom, the switches in the middle, and the router at the top (Figure 5-53).
6. Use the **Copper Straight-Through** connection to connect each of these devices as described next:
   a. On each workstation, connect the Ethernet cable to the **FastEthernet0** interface.
   b. On each switch, connect the Ethernet cable from the workstation to the **FastEthernet0/1** interface. Connect the Ethernet cable to the router to the **FastEthernet0/2** interface.
   c. On the router, connect Switch0 to the **FastEthernet0/0** interface, and connect Switch1 to the **FastEthernet1/0** interface.
   d. Wait a few minutes for the workstation-to-switch connections to turn to green triangles on both ends of each connection.

Figure 5-53 Arrange the devices in a pyramid shape



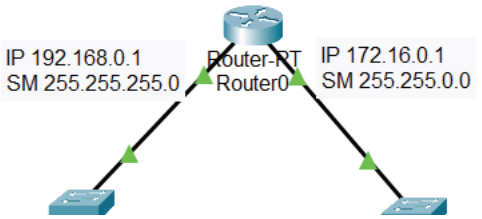
Source: Cisco Systems, Inc.

The router and the switch must be configured for the connections to come up:

7. Click **Router0** to open its configuration window. Click the **Config** tab. As you make changes, notice the commands that appear in the Equivalent IOS Commands pane at the bottom of the window.
8. In the left pane, click the **FastEthernet0/0** interface. Make the following changes to the interface's configuration:
   - IP Address: **192.168.0.1**
   - Subnet Mask: **255.255.255.0**
   - Port Status: **On**
9. Click the **FastEthernet1/0** interface. Make the following changes to the interface's configuration:
   - IP Address: **172.16.0.1**
   - Subnet Mask: **255.255.0.0**
   - Port Status: **On**
10. Close the **Router0** window and wait a few minutes for the switch-to-router connections to turn to green triangles on both ends of each connection.

11. Click **PC0** to open its configuration window. Click the **Desktop** tab and then click **IP Configuration**. Make the following changes:
    - IP Configuration: **Static**
    - IP Address: **192.168.0.100**
    - Subnet Mask: **255.255.255.0**
    - Default Gateway: **192.168.0.1**
12. Close the IP Configuration window without closing the full configuration window by clicking the small **X** on the blue bar near the upper right corner. Then click **Command Prompt**. Enter `ipconfig` to confirm the network configuration is correct.
13. Close **Command Prompt** and close the **PC0** configuration window.
14. Click **PC1** to open its configuration window. Click the **Desktop** tab and then click **IP Configuration**. Make the following changes to the workstation's configuration:
    - IP Configuration: **Static**
    - IP Address: **172.16.0.100**
    - Subnet Mask: **255.255.0.0**
    - Default Gateway: **172.16.0.1**
15. Close the IP Configuration window by clicking the small **X** on the blue bar near the upper right corner. Then click **Command Prompt**. Enter `ipconfig` to confirm the network configuration is correct.
16. Enter the command `ping 192.168.0.100`. Was the ping successful? If so, then you have successfully begun building your Packet Tracer network. If not, troubleshoot your network to determine where the problem is and fix it. What problems did you have to fix, if any?
17. In the main Packet Tracer window, open the **Simulation Panel** (click the **Simulation** button in the lower-right corner, or press **Shift+S**). In the PC1 configuration window, run the ping again. Move or minimize the PC1 configuration window so you can see the devices in the workspace. In the Simulation Panel, click the **Play** button and watch what happens to the devices in your workspace.
18. When you're ready, click any ICMP message in the Event List to examine its details and to see an explanation of each step in the process. Click through the various layers and read their explanations. Click the Inbound PDU Details and Outbound PDU Details tabs to examine the headers of each ICMP message. Explore at least three different ICMP messages and respond to the following questions for each of the three messages:
    a. Which two devices are exchanging each message (sender and receiver)?
    b. Which OSI layers added headers to each message?
19. When you've explored the ICMP messages, in the Simulation Panel, click the **Reset Simulation** button to stop the simulation. Close the **Simulation Panel**, close **Command Prompt**, and close the **PC1** window.
20. Add a **Note** to each connection that lists its IP address, subnet mask, and default gateway if relevant. Figure 5-54 gives an example for the router, which has two interfaces with IP addresses. After adding all your documentation, **take a screenshot** of your Packet Tracer network; submit this visual with your responses to this project's questions.

Figure 5-54 Good documentation makes troubleshooting easier



Source: Cisco Systems, Inc.

21. Click **File** and **Save**. Give the file an informative name (such as NetPlusPTnetwork), and save your Packet Tracer file in a safe place for future projects. What is the name of your Packet Tracer file?
22. Add the Packet Tracer network file's name and location to the Packet Tracer page on your Wikidot website, along with any notes you think might be helpful to you for the next Packet Tracer project. When you're finished, close Packet Tracer.

Note to Instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors site. Some Packet Tracer projects build on earlier Packet Tracer networks. If needed for one or more students, you can provide a previous project's solution file as a start file for one of these progression projects.

Capstone Project 5-2: Explore IOS Command Modes in Packet Tracer

Estimated time: 30 minutes
Objective 5.5: Given a scenario, use the appropriate tool or protocol to solve networking issues.
Resources:
- Computer with Cisco Packet Tracer installed

Description:
This Packet Tracer project starts with a new network and is *not* intended to build on your Packet Tracer network for the Capstone Projects. For this project, do *not* use your existing Packet Tracer network that you just created in Capstone Project 5-1—you'll come back to that network file in later chapters. In this project, start with a new Packet Tracer file to experiment with IOS command modes. You will *not* need to save this network for future projects or chapters.

The command modes in Cisco's IOS (Internetwork Operating System) provide structure to the types of commands you can use as well as offering the opportunity to define hierarchical access control to networking devices. For example, you might want to give a new technician the ability to see routes on a router but not to make configuration changes to the router. With command modes, you can configure different passwords at different levels to manage this access (which you'll do in a later chapter). It can be confusing at first to remember which modes allow which commands.

Different commands are available depending on the mode you're in. For example, when you first start working with a router, you begin in the user EXEC mode. To step up to the privileged EXEC mode, you enter enable, which can also be abbreviated as simply en. Table 9-4 explains the most used modes on Cisco routers and switches.

To take one step down from a higher mode into a lower mode, enter the **`exit`** command. From any higher mode, enter the **end** command or press Ctrl+Z to return to privileged EXEC mode.

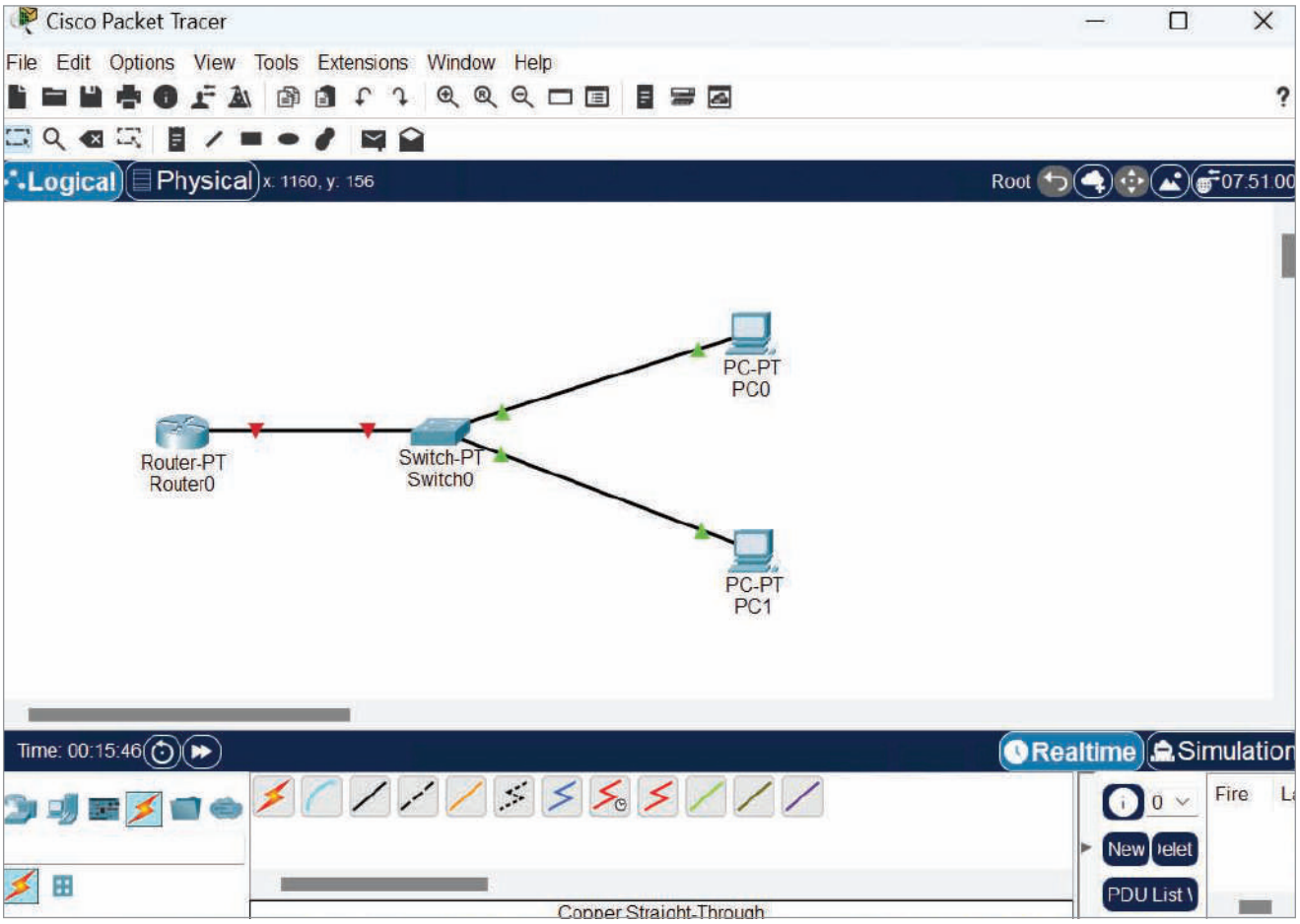
To explore commands available in various modes on Cisco switches and routers, complete the following steps:

1. Open Cisco Packet Tracer and, if necessary, sign in with your Skills for All account.
2. Add the following devices to your workspace (Figure 5-55):
   a. Two **PCs**
   b. One **PT-Switch**
   c. One **PT-Router**
3. Connect all the devices as illustrated in Figure 5-55.

Table 9-4 Cisco CLI modes

Mode	Default prompt on a router	Command to enter mode	Description
User EXEC	`Router>`	Login to device	Offers limited commands to evaluate current configurations without making changes to the configuration.
Privileged EXEC	`Router#`	`enable` or `en`	Typically requires a password and offers access to all EXEC commands, which provide tools for testing and helpful information about the current configurations. You can also run EXEC commands from other modes by prefacing the EXEC command with the **do** command.
Global configuration	`Router(config)#`	`configure terminal` or `conf t`	Allows device configuration changes and gives access to more specific configuration modes.
Interface configuration	`Router(config-if)#`	`interface` or `int`	Allows configuration changes to an interface. Specific configuration modes for features or protocols allow changes to configurations such as interfaces, DHCP, and routing. Configuration submodes and subsubmodes dig deeper into configuration options.

Figure 5-55 Network for exploring IOS modes



Source: Cisco Systems, Inc.

4. Configure the router's interface with the following information:
   - IP Address: **192.168.10.1**
   - Subnet Mask: **255.255.255.0**
   - Port Status: **On**
5. Configure PC0 with the following information:
   - IP Configuration: **Static**
   - IP Address: **192.168.10.10**
   - Subnet Mask: **255.255.255.0**
   - Default Gateway: **192.168.10.1**
6. Configure PC1 with the following information:
   - IP Configuration: **Static**
   - IP Address: **192.168.10.20**
   - Subnet Mask: **255.255.255.0**
   - Default Gateway: **192.168.10.1**
7. Document your configurations in your workspace. When you're finished, **take a screenshot** of your network with its IP addressing documentation; submit this visual with your responses to this project's questions.
8. Confirm that each PC can ping its default gateway. What command did you use? If either ping fails, troubleshoot and fix the problem. What corrections did you have to make?
9. Confirm that each PC can ping the other PC. What commands did you use? If either ping fails, troubleshoot and fix the problem. What corrections did you have to make?
10. Open Switch0's configuration window, and then click the **CLI** tab. You might need to press **Enter** to get a prompt. What mode are you in when you first access the CLI? How do you know?
11. Enter `?` to display more information. What commands can you enter in this mode?
12. Enter `enable`. What is the prompt now? What mode is this?
13. Enter `show running-config` (or `sh run`) to view the running configuration. What is the switch's current hostname?
14. Press **Enter** to move forward one line at a time. Press **Space** to move forward one page at a time and return to the prompt.
15. Enter `configure terminal` (or `conf t`) to enter global configuration mode. What is the prompt now?
16. Enter `hostname Stampy` to change the hostname. What is the prompt now?
17. Without changing modes, you can list the switch's physical and data link interface information. Enter the command `do show interface` (or `do sh int`).
18. Scan through the command's output. Which interfaces are up?
19. Enter the `interface` command followed by the name of one interface that is up to change to interface configuration mode for that interface. For example, you might enter `interface fastethernet0/1` (or `int fa0/1`) if the Fast Ethernet 0/1 interface is up on your switch (meaning you connected a device to it). Make sure to use the name of an interface that is up on your switch.
20. What is the prompt now? What mode are you in?
21. You are not required to save this Packet Tracer network from this project, but you might want to save it for future reference or testing. Make some notes on your Wikidot website about your activities in this project.

Note to Instructors: A Packet Tracer solution file is provided for each Packet Tracer project through the Instructors site.

Solutions to Self-Check Questions

Section 5-1: Transmission Basics

1. When you measure the amount of data coming into your home network, what metric are you identifying?
   Answer: d. Throughput
   Explanation: **Throughput** refers to the number of data bits (0s and 1s) that are actually received across a connection each second. "Bandwidth" refers to the amount of data you could theoretically transmit during a given period. Noise, or interference, degrades or distorts a signal on a network. Duplex indicates if and when a signal can travel in both directions on a medium.

2. Which of the following measures the change in signal strength between transmission and reception?
   Answer: b. Decibel
   Explanation: The change in signal strength between transmission and reception is measured in **dB (decibels)**. Latency is the delay between when data leaves the source and when it arrives at its destination. If packets experience varying amounts of delay, they can arrive out of order—a problem commonly called jitter. Crosstalk occurs when a signal traveling on one wire or cable infringes on the signal traveling over an adjacent wire or cable.

Section 5-2: Copper Cable

3. What is the minimum twisted-pair category required for 10-Gigabit Ethernet at 100 meters?
   Answer: c. Cat 7
   Explanation: **Cat (Category) 7** and Cat 8 both support throughput of 10 Gbps or higher at the full 100 meters segment length. Cat 6a can support throughput of 1 Gbps or, at shorter distances, 10 Gbps. Cat 5e maxes out at 1 Gbps.

4. Pin 1 on one end of a cable is orange and white striped. What color should pin 1 be on the other end of the cable to create a crossover cable?
   Answer: d. Green and white striped
   Explanation: A crossover cable reverses the transmit and receive wires. If pin 1 on one end of a cable is orange and white striped, then the other end of the crossover cable would need to cross the orange and white striped wire to pin 3. Pin 1, then, would need to receive the **green and white striped wire** from pin 3 on the first end. If the orange and white striped wire continues to pin 1 on the second end, the cable is more likely a straight-through cable. The solid blue wire typically stays in the middle at pin 4 or possibly crosses over to pin 7. The solid brown wire typically stays on the other end of the connector at pin 8 or possibly crosses over to pin 5.

5. What is the typical maximum cable segment length for Ethernet networks?
   Answer: b. 100 meters
   Explanation: The maximum cable segment length for both STP and UTP is **100 meters**, or 328 feet, on Ethernet networks that support data rates from 1 Mbps to 10 Gbps. Some categories of STP require shorter segment lengths to achieve maximum throughput, which can vary from 10 Mbps to 100 Gbps.

Section 5-3: Fiber-Optic Cable

6. Which of the following statements is true?
   Answer: b. SMF supports higher bandwidths than MMF.
   Explanation: Because it reflects little, the light in an SMF (single mode fiber) cable does not disperse as the signal travels along the fiber. This continuity allows SMF to accommodate the highest bandwidths and longest distances (without requiring repeaters) of all network transmission media. Therefore, **it's true that SMF supports higher bandwidths than MMF (multimode fiber)**.

SMF's core is 8 to 10 microns in diameter, which is thinner than MMF's core at 50 to 62.5 microns. MMF is less expensive to install and, therefore, typically used to connect routers, switches, and servers on the backbone of a network within a data center; because of its relatively high cost, SMF is rarely used for such short connections. Many pulses of light generated by a laser or LED light source travel at various angles through MMF's larger core, thus bouncing off of (that is, reflecting) the interior surfaces of the core and, therefore, suffer from greater attenuation than SMF signals. Inside the SMF core, laser-generated light travels a single path over the core, reflecting very little. Because it reflects little, the light does not disperse as the signal travels along the fiber.

7. Which fiber connector contains many strands in a single ferrule?
Answer: b. MPO
Explanation: The **MPO (multi-fiber push on)** connector typically contains 8, 12, or 24 strands of fiber in a single ferrule, and special use case cables might contain 32, 48, 60, or even 72 fibers in each ferrule. SC (subscriber connector) and LC (local connector) fiber connectors each contain a single fiber and are available in full-duplex mode by pairing two ferrules. ST (straight tip) connectors contain a single fiber in the connector and are not available in full-duplex mode.

8. What is the earliest transceiver type to support four channels on a single transceiver to increase throughput?
Answer: a. QSFP
Explanation: **QSFP (quad small form-factor pluggable)** transceivers squeeze four channels in a single transceiver to support data rates up to 40 Gbps (4 × 10 Gbps). QSFP+ improved on the same technology to support even higher data rates. Older transceivers, such as SFP (small form-factor pluggable) and SFP+, do not support multiple channels per transceiver.

Section 5-4: Cable Troubleshooting Tools

9. Which tool could you use to determine if a laptop's power cable is working properly?
Answer: d. Multimeter
Explanation: You might use a **multimeter** to measure voltage to verify that a cable is properly conducting electricity, whether that cable is a data cable on a network or a power cable. A cable performance tester performs sophisticated calculations on a data signal to determine many characteristics of a data cable. A continuity tester determines whether a cable is carrying a data signal to its destination. A toner and probe kit can help determine where a wire, possibly out of hundreds, terminates.

10. Which tool could you use to test a twisted-pair cable's pinout?
Answer: d. Continuity tester
Explanation: Some **continuity testers** will verify that the wires in a UTP or STP cable are paired correctly following TIA/EIA 568 standards. OTDRs and OPMs are used to test fiber-optic cables. A toner probe can help determine where a wire, possibly out of hundreds, terminates.